Singha Junchan

Security Engineering: Cloud Security · Identity & Access (IAM) · AI-Agent Security · Governance & Risk

web portfolio

singhaju.github.io | github.com/singhaju | linkedin.com/in/singha-junchan | Thailand

Profile

Security-minded computer engineer who believes good security should enable a business to move forward with confidence. Driven by challenge (1st in Thailand at TOI18, after more than 1,000 hours of training), quick to adapt and collaborative: built AI agents for third-party risk management at Allianz Ayudhya, leads both the SIIT student committee and the development of Thammasat University's incident-reporting system, and works on eleven research projects in cloud, AI and post-quantum security, one with a Best Paper Award. Seeking internships from January 2027 and full-time roles from May 2027.

Education

B.Eng. in Computer Engineering, specialisation in Cloud Computing & Cybersecurity — Aug 2023 – May 2027 (expected)
Sirindhorn International Institute of Technology (SIIT), Thammasat University — *GPA 3.67 / 4.00*

First-class honours expected; full scholarship (4-Year Continuing Academic Outstanding Student Program).

Experience

Information Security Intern – *Allianz Ayudhya, Bangkok* — Jun – Jul 2026

- Built two AI agents that take on the routine first pass of third-party risk management, until then a largely manual review, while people keep every decision; both were tested with the security team on real, anonymised cases.
- One drafts answers to partners' due-diligence questionnaires with evidence, a rationale and a confidence score, and proposes a remedy for every gap; the other guides business owners through each step and flags missed ones.
- Learned governance and risk management from the CISO, within ISO 27001, NIST and internal policy.

Tech Lead – *TU Alert, built for Thammasat University, launching soon* — 2026 – Present

- Lead the student team hired to build the university's incident-reporting system, and write code (TypeScript, Bun, Next.js): students report through LINE with photos and a map location; each report reaches the office responsible.
- Designed the architecture and the security: two-factor staff sign-in, personal-data retention and erasure, Semgrep and gitleaks in CI.

R&D Intern, AI Research Team – *Amity Solutions, Bangkok* — Jul – Aug 2024

- Turned recent LLM research into retrieval-augmented (RAG) chatbots that answer from a knowledge base.

Research

Eleven projects with Prof. Somchart Fugkeaw (SIIT) on one question: how to make strong security checks cheap enough to run on every request, and provable afterwards: 2 accepted, 3 submitted, 6 in preparation.

- **U-HAP** (Best Paper Award, IEEE JCSSE 2026, second author): one Kubernetes policy language for ACL, RBAC, ABAC and deny rules, compiled into indexes; matched OPA decision for decision while applying updates faster.
- **FEDAC** (first author, submitted to IEEE TDSC): zero-trust access that verifies each caller once, up to a zero-knowledge proof, then lets a sidecar enforce a short-lived capability locally (Go, Rust, AWS EKS).
- **ZKAudit, ZK-ALBA, PQZ-Audit** (senior project; ZKAudit accepted, KST 2027): auditors prove credentials and scope in zero knowledge; PQZ-Audit is designed to check answers over encrypted logs without decryption.
- **PQ-SCVA-MCK** and **SWARM-Fog** (first author, submitted): post-quantum verified trust across Kubernetes clusters; self-healing orchestration for industrial IoT with no central controller.
- *In preparation:* MoCA (AI-agent containment), Fed-RAG-Hunt (federated threat hunting), encrypted SIEM, freshness-aware multi-cloud scaling.

Hands-on Security

Ethical Hacking and Digital Forensics labs, SIIT 2026 · SOSECURE Road to Cybersecurity 2025 (Linux hardening, web pentesting, SOC) · NCSA CTF Thailand Tournament 2026 and Hack The Scammer CTF 2025 · home lab (Kali Linux), TryHackMe, Hack The Box

Leadership & Teaching

- **President**, Sub Student Committee 32, SIIT (since 2024): organised 11 student events and led SIIT First Meet 2024, welcoming about 450 first-year students with about 200 student organisers.
- **Founder & Lead Instructor**, TOI Tutor and ABIG Center (since 2022): coach students for Thailand's programming olympiads: hundreds of students, self-written courses.

Skills

Security & IAM:	Zero Trust, OIDC/JWT, MFA (TOTP), RBAC/ABAC, OPA/Rego, Kubernetes authorization
Governance & Risk:	ISO 27001, NIST, third-party risk management, due-diligence questionnaires, gap analysis
Cloud & AI:	AWS (EKS, EC2), Kubernetes, Docker, Azure basics; LLM agents, RAG, MCP, prompt-injection testing
Offensive & SOC:	Nmap, Wireshark, Burp Suite, Metasploit, pwntools, hashcat, OWASP Top 10; Splunk, Wazuh, Suricata
Cryptography:	zk-SNARKs (Groth16), Merkle commitments, post-quantum ML-DSA and ML-KEM
Programming:	Python and C++ (main); Go, Rust, TypeScript, C, Java, Bash, SQL
Languages:	Thai (native), English (fluent), Chinese (basic) · **Interests:** Muay Thai, board games, food and travel

Awards & Certifications

- **1st in Thailand**, national champion and gold medallist: Thailand Olympiad in Informatics (TOI18), 2022.
- **Best Paper Award**, IEEE JCSSE 2026 · 1st place (team), Image Recognition Challenge (Osaka Inst. of Technology, SIIT), 2025.
- ISC2 Certified in Cybersecurity (CC), 2025 · CompTIA Security+ (exam planned Dec 2026).

References

Apichai Wongchindakhun (CISO, Allianz Ayudhya; internship mentor) | linkedin.com/in/apichai-wongchindakhun-1839b314
Prof. Somchart Fugkeaw (SIIT, Thammasat University; research supervisor) | linkedin.com/in/somchart-fugkeaw-a147b9111
Pachara Wongthanakarn (TU Alert co-developer) | linkedin.com/in/pachara-wongthanakarn